MIL-S-12646A (Ord)

6 April 1955
SUPERSEDING
MIL-S-12646 (Ord)
16 April 1953

MILITARY SPECIFICATION
SIGHT, COMPUTING, M19A1

1. SCOPE

1.1 This specification covers a computing sight, for use on Gun, 40MM, M1A1 on Carriages, Gun, AA., 40MM, M2A3 and M27.

2. APPLICABLE DOCUMENTS

2.1 The following specifications and drawings of the issue in effect on the date of invitation for bids, form a part of this specification:

SPECIFICATIONS

Military

MIL-F-13926	Fire Control Materiel; General Specification Governing the Manufacture and Inspection of

DRAWINGS

Ordnance Corps

F7647810	Sight, Computing, M19A1

(Copies of specifications, standards, drawings, and publications required by the contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer).

3. REQUIREMENTS

3.1 General.- Unless otherwise specified, the requirements specified in Specification MIL-F-13926 which are common to all fire control instruments shall be applicable to this specification.

FSC 1220

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3.2 Material.- Material shall be in accordance with the applicable drawings and specifications forming a part of this specification.

3.3 Adjustment

3.3.1 Elevation linkage.- The parallelogram linkage Support D7647819, with pads for quadrant, shall repeat the simulated gun elevation on the test stand within plus or minus 1 mil, after it has been adjusted in accordance with 4.2.2 (a).

3.3.2 Boresighting.- Telescope adjustments shall be set to within plus or minus 1 mil of alinement with the boresight mark on the test target.

3.3.3 Superelevation.- Superelevation shall be set to 9 mils within plus or minus 1 mil at zero elevation in accordance with 4.2.2 (j).

3.3.4 Bail brake.- The force required to move the bail shall not exceed 40 pounds for any position of the bail between 65 degrees dive and 55 degrees climb after it has been adjusted in accordance with 4.2.2 (k).

3.4 Performance

3.4.1 Limits of operation.- The computing sight shall operate satisfactorily between the limits of minus 6 degrees and plus 90 degrees simulated gun elevation on the test fixture.

3.4.2 Problem accuracy.- The sight shall compute deflection to an accuracy of 5 mils average error radially (see 4.3.1.1), except at zero speed with zero elevation and plus 1600 mils angle of approach, where the accuracy shall not exceed 2 mils average error radially. At least 80 percent of the average errors shall fall within the required tolerance and no reading at any check point shall exceed an error of 10 mils radially. The "backlash figure of merit" (see 4.3.1.1) shall not exceed 4.5 mils.

3.4.3 Backlash

3.4.3.1 Vertical.- Vertical backlash measured at the telescope shall not exceed 2 mils at 700 miles per hour, zero climb and dive, and plus 1600 mils angle of approach, when tested in accordance with 4.3.2.1.

3.4.3.2 Azimuth.- Azimuth backlash measured at the telescope shall not exceed 4 mils at zero miles per hour, zero climb and dive, and zero mils angle of approach and shall not exceed 6 mils at 700 miles per hour, zero climb and dive, and plus 1600 mils angle of approach, when tested in accordance with 4.3.2.2.

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3.4.3.3 Slide.- The backlash measured at the base of Slide C7675373 shall not exceed 0.015 inch, total indicator reading, at zero speed, zero climb and dive, and zero angle of approach, when a force of 5 pounds is alternately applied in opposite directions at the end of the slide.

3.4.3.4 Handwheel.- Backlash between the course handwheel and the azimuth indicating scale on the computer head shall not exceed 5 degrees measured at the course handwheel.

3.4.4 Torque

3.4.4.1 Shaft.- With the computing sight and fixture set at zero elevation, the torque required at Shaft B7635573 to operate the sight in clockwise and counterclockwise directions shall not exceed 25 ounce inches when measured at angle of approach settings of zero mils, plus 1600 mils, plus 3200 mils, and minus 1600 mils, with zero speed and zero inclination settings.

3.4.4.2 Handwheel.- The torque required at Handwheel Assembly C7675852 to rotate the computer head shall be between 18 and 25 pound inches.

3.4.5 Tape tension.- Tape tension shall provide a torque between 50 and 80 pound inches at the telescope, when measured by applying torque in a clockwise direction for 5 degrees movement from a reference point.

3.4.6 Air speed knob.- With the angle of approach set at plus 800 mils, dive and climb angle set at 40 degrees dive, and gun elevation set at zero, the torque required to rotate the air speed knob shall be between 15 and 45 pound-inches, when measured in accordance with 4.4.2.2 (1).

3.5 Environmental

3.5.1 Temperature.- The computing sight shall meet the requirements specified in 3.4.2 throughout the air temperature range extending from plus 51.5°C (+ 125°F) (minimum exposure of 4 hours with full impact of solar radiation, 360 BTU/Ft. Sq/Hr.) to minus 40°C (- 40°F) (minimum exposure of 3 days without benefit of solar radiation). The computing sight shall be capable of safe storage throughout the air temperature range extending from plus 71°C (+ 160°F) (for periods as long as 4 hours per day) to minus 62.2°C (- 80°F) (for periods of at least 3 days duration). Test shall be in accordance with 4.4.2.

3.5.2 Humidity.- The computing sight shall not suffer damage or fail to meet the requirements specified in 3.4.2 immediately after being subjected to the humidity test specified in 4.4.3.

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3.5.3 Rain.- The computing sight shall show no evidence of water penetration and shall not suffer damage immediately after being subjected to the rain test specified in 4.4.4.

3.5.4 Vibration.- The computing sight shall not suffer damage or fail to meet the requirements specified in 3.4.2 immediately after being subjected to the vibration test specified in 4.4.5.

3.5.5 Shock.- The computing sight shall not suffer damage or fail to meet the requirements specified in 3.4.2 immediately after being subjected to the shock test specified in 4.4.6.

3.6 Workmanship.- Workmanship shall be in accordance with Specification MIL-F-13926.

4. QUALITY ASSURANCE PROVISIONS

4.1 General.- Unless otherwise specified each computing sight shall be inspected for conformance with this specification and with Specification MIL-F-13926. Unless otherwise specified, tests shall be performed at an ambient temperature between 60°F and 90°F, a barometric pressure between 28 and 32 inches of mercury, and a relative humidity not greater than 80 percent.

4.2 Adjustments

4.2.1 Equipment required

4.2.1.1 Test target.- The test target shall accommodate maximum sight deflection for a target distance of not less than 18 feet. Layout of check points shall be in accordance with "Test Target Layout Parameters", shown in Appendix of this specification.

4.2.1.2 Test fixture.- The test fixture shall simulate the gun breech casing with respect to mounting and shall provide a means for introducing elevation into the sight between the limits of minus 6 degrees and plus 90 degrees. The centerline of the test fixture shall be level and perpendicular within 0.5 mil to a like centerline on the test target. The fixture also shall contain a power device capable of driving the computer head in azimuth, in both a clockwise and counterclockwise direction at 5±1 degree per second and 45 ± 2 degrees per second.

4.2.1.3 Telescope.- Since the reflex sight has a significant inherent parallax when used in conjunction with a close test target, a straight tube telescope of the M7 type shall be used in making the necessary tests and adjustments. An adapter shall be used for mounting the telescope so that the height of the optical axis above the center of rotation shall be 6.186 inches.

4.2.2 Procedure.- With the sight set upon the test fixture, the following adjustments and checks shall be made:

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(a) With the test fixture at zero elevation, set angle of approach at zero mils, dive and climb bail in a vertical position and speed knobs against the zero stops. Adjust Linkage A7591401 until a zero reading is obtained on a zeroed gunner's quadrant placed on the Support D7647819. Secure locknuts in position. Place a zeroed gunner's quadrant on the machined pads of Body Assembly D7647816 and adjust Stem B7648328 of the ball joint assembly at the bottom of the computer head to obtain a zero reading on the quadrant.

(b) Set test fixture at 1600 mils elevation and all other conditions same as (a) above. Place an adapter and a zeroed gunner's quadrant on the reflex sight mounting stud and check for a zero reading on the quadrant. If necessary, adjust the elevation adjusting worm of the support assembly until zero reading is obtained.

(c) With the test fixture at 1600 mils elevation, angle of approach at approximately 1600 mils, and all other conditions same as (a) above, place a zeroed gunner's quadrant on the machined pads of Body Assembly D7647816 and then on the quadrant adapter. The reading should be the same, zero for both. If necessary, adjust Rod A7591423 until zero readings are obtained. Maintain this condition except change the air speed knob setting to 700 miles per hour; the angle induced at the machined pads of Body Assembly D7647816 shall be reproduced at the reflex mounting stud within 1 mil when measured with a zeroed gunner's quadrant.

(d) Check tape tension. It shall meet the requirements of 3.4.5.

(e) With the test fixture at zero elevation, set angle of approach at approximately plus 1600 mils. Maintain position of dive and climb bail and air speed knobs as set in (a) above. Boresight the telescope to a reference point on the target. Power rotate the computer head at 5 $\pm$ 1 degree per second and 45 $\pm$ 2 degrees per second in both clockwise and counterclockwise directions. Movement of the telescope vertical reticle line shall not exceed an excursion of 4 mils when viewed on the target for each of the rates indicated when checked in each direction of rotation. If excursion of more than 4 mils is visually apparent it may be necessary to make adjustments to the stop screws in the air speed Knob Assembly C7652981.

(f) With the test fixture at 1600 mils elevation position set the angle of approach at zero mils. Adjust the angle of approach until the line of sight through the telescope remains on the horizontal reference line

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within 0.5 mil, while the speed knob is rotated from 0 to 700 miles per hour. Set the angle of approach index to coincide with the zero graduation on the computer head.

(g) With the test fixture at zero elevation, set angle of approach to plus 1600 mils. Rotate the speed knob from 0 to 700 miles per hour, while adjusting the position of the computer head in a horizontal plane until the line of sight of the telescope remains on the target reference line within 0.5 mil.

(h) With the test fixture at zero elevation, set angle of approach at 3200 mils. Adjust the climb and dive bail until the line of sight of the telescope follows a horizontal reference line within 1 mil when the speed knob is rotated from 0 to 700 miles per hour. Set the bail indices to zero. Repeat at zero mils angle of approach.

(i) With the test fixture at zero elevation, set angle of approach at plus 1600 mils and the speed knob at 700 miles per hour. Move the bail through the full length of travel. The line of sight of the telescope shall remain on a vertical reference line within 1 mil.

(j) Introduce superelevation into the sight as follows: With the test fixture at zero elevation, set the angle of approach to plus 1600 mils. Set speed knob at boresight mark (BS) and set in 50 degrees dive angle. Place a zeroed gunner's quadrant on the machined pads on Body Assembly D7647816 and obtain a zero reading by adjusting Stem B7648328 at the bottom of the computer head. The telescope will now be alined on the boresight mark. Set the speed knob at zero speed, and dive bail to zero. Superelevation is now set into the sight. Observe the line of sight; it shall meet the requirements specified in 3.3.3. Introduce a 50 degrees dive angle, and set speed knob to boresight mark (BS). This will remove superelevation. The line of sight now should be on the boresight mark of the test target.

(k) Bail brake adjustment shall be set so that when the speed knob is rotated slowly through the entire range individual clicks are felt and counted. The bail shall not move more than the width of a graduation line on the dive and climb scale when the speed knob is rotated. The force required to move the bail, measured with a spring scale applied to the bail at the course arrow, shall be in accordance with 3.3.4. The bail efforts

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are to be determined with the speed set at 600 mph, angle of approach set at plus 800 mils, and gun elevation set at zero.

(1) Air speed knob torque shall be measured at 200 mph, 400 mph, and 600 mph with the computer course, dive, and elevation set in accordance with 3.4.6. Torque measurements shall be made by repeated 20 mph reversals of the air speed knob. The first 2 reversals for a given speed setting shall not be measured.

4.3 Performance

4.3.1 Problem accuracy.- The computing sight settings and simulated gun elevation for each check point shown in the Appendix shall be introduced into the sight. The line of sight of the telescope shall then fall on the respective check points of the test target, within the required tolerance.

4.3.1.1 Accuracy readings shall be read using both the increasing and decreasing angle of approach. The "average error" is defined as one-half the sum of both radial readings. The "backlash figure of merit" is defined as the sum of the computed "average errors" divided by the number of check points at which readings are taken.

4.3.2 Backlash

4.3.2.1 Vertical.- The speed shall be set at 700 miles per hour, angle of approach at plus 1600 mils, and climb and dive at zero. The telescope shall then be set on target, using hand pressure, and test shall then be made as follows:

(a) Gently apply and remove a torque of 6 pound inches about the horizontal axis of the telescope holder and note the line of sight at the target.

(b) Reverse the direction of torque applied in (a) and note the line of sight. The difference between the readings obtained in (a) above and (b) is vertical backlash and shall meet the requirements specified in 3.4.3.1.

4.3.2.2 Azimuth.- The speed shall be set at zero miles per hour, angle of approach at zero mils, and climb and dive at zero. The telescope shall be set on target, using hand pressure, and test shall be made as follows:

(a) Gently apply and remove a torque of 6 pound inches about the vertical axis of the resolution mechanism and note the deflection of the line of sight on the target.

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(b) Reverse the direction of torque applied and note the deflection. The difference between the readings obtained in (a) above and (b) is computer and resolution mechanical backlash and shall meet the requirements specified in 3.4.3.2. Repeat test, setting speed at 700 miles per hour, angle of approach at plus 1000 mils, and climb and dive at zero. Backlash shall meet the requirements specified in 3.4.3.2.

4.4 Environmental

4.4.1 General.- The environmental tests specified shall be limited to those contained in the following sub-paragraphs. Three of the first five computing sights and one out of every additional 100 or fraction thereof shall be tested.

4.4.2 Temperature.- The computing sight shall be exposed to an air temperature of minus 62.2°C (-80°F) for a period of 3 days. The air temperature shall then be raised to minus 40°C (-40°F) and after the internal temperature of the unit has reached equilibrium, (approximately 4 hours), it shall be tested for conformance with 3.5.1. The air temperature shall be raised to plus 71°C (+160°F) and after the internal temperature of the unit has reached equilibrium at that temperature, it shall be allowed to remain at that temperature for a period of 4 hours. The air temperature shall be reduced to plus 51.5°C (+125°F) and after the internal temperature of the unit has reached equilibrium at that temperature, the top surface shall be exposed to 360 BTU/Ft. Sq/Hr., for a period of 4 hours. The unit shall then be tested for conformance with 3.5.1.

4.4.3 Humidity.- A test chamber that is capable of being sealed shall be used; the source of heat shall be arranged in such a manner that the radiant heat will not fall upon the computing sight being tested. A unit shall be placed in the chamber, the temperature of which is adjusted to plus 46.1°C ± 1.1°C (+115°F ± 2°F), and the relative humidity adjusted to 95 ± 5 percent. The unit shall remain under the above conditions for a period of 6 hours. The temperature of the chamber shall be adjusted to plus 26.7°C ± 1.1°C (+ 80°F ± 2°F) and the unit shall be allowed to remain under this new condition for a period of 6 hours. The transitional period shall be 2 hours; the rate of change during that period shall be uniform. The above cycle shall be repeated 10 times. Immediately after this test the unit shall be tested for conformance with 3.5.2.

4.4.4 Rain.- The computing sight shall be mounted in the test chamber to simulate installation conditions. The rain test chamber temperature shall be maintained between 20°C and 30°C (68°F and 86°F) throughout the test period. A simulated rainfall of 4 + 1 inch per hour shall be produced by a suitable means of such design that the water is emitted in the form of small droplets rather than a fine mist.

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The temperature of the water shall be maintained between 11°C and 24°C (51.8°F and 75.2°F). The water spray shall be directed on the computing sight for a period of 2 hours in directions ranging from vertical to 45° from the vertical. The simulated rain shall be directed on all surfaces of the computing sight that are normally subjected to rain when the instrument is in its operating position. Immediately after this test the unit shall be tested for conformance with 3.5.3.

4.4.5 Vibration.- The computing sight shall be mounted using its own mounting provisions, to simulate field use. Each unit tested shall be vibrated successively along each of the three mutually perpendicular directions with accelerations and excursions as shown below. Applied frequency shall be changed at a uniform rate between the limits indicated; the rate of change shall not exceed 1 cycle per second. Immediately after this test the unit shall be tested for conformance with 3.5.4.

Frequency limits	Excursion	Acceleration	Time (each direction)
0 - 10 cps	0.060 inch	- - - - -	1 hour
10 - 20 cps	0.030 inch	- - - - -	2 hours
20 - 400 cps	- - - -	2.5 g's	2 hours

4.4.6 Shock.- The computing sight shall be mounted rigidly using its own mounting provisions to simulate field use. A shock of 20 g's shall be imparted three times in each direction along each of the three mutually perpendicular directions making a total of 18 shock impulses. Each shock impulse shall have a time duration of 11 ± 1 milliseconds; maximum g value shall occur at approximately 5.5 milliseconds and shall be within ± 10 percent. Filters used shall have cut-off frequencies selected to eliminate all harmonics above the fundamental from the measured shock impulse. Immediately after this test the unit shall be tested for conformance with 3.5.5.

5. PREPARATION FOR DELIVERY

5.1 Packaging, packing and marking.- Level of packaging, packing and marking shall be in accordance with instructions forming a part of the contract or furnished by the contracting officer in connection with the contract.

6. NOTES

6.1 Ordering data.- Contracts and invitations for bids should specify the title, number, date of this specification and level of packaging and packing.

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APPENDIX

Simulated Gun Elevation	Computing Sights Settings			Test Target Layout Parameters			Check Points
	Dive Angle	Air Speed (MPH)	Approach Angle-Mils	L/b	V/b	E_v Inches	
		200	0	.1194	-.0090	0	1
		200	3200	-.1194	-.0090	0	2
Zero	Zero	400	0	.2388	-.0090	0	3
		400	3200	-.2388	-.0090	0	4
Mils	Degrees	700	0	.4179	-.0090	0	5
		700	800	.4194	-.0128	0	6
		700	2400	-.4194	-.0128	0	7
		700	3200	-.4179	-.0090	0	8
		360	0	.2149	-.0064	0	9
		360	1600	0	.1717	.08	10
300	Zero	360	3200	-.2149	-.0064	0	11
		360	-1600	0	-.1374	.06	12
Mils	Degrees	700	0	.4179	-.0064	0	13
		700	1600	0	.4104	.46	14
		700	3200	-.4179	-.0064	0	15
		700	-1600	0	-.2330	.17	16
		700	0	.2955	.2865	.24	17
Zero	45	700	800	.2641	.3621	.37	18
Mils	Degrees	700	2400	-.2641	.3621	.37	19
		700	3200	-.2955	.2865	.24	20

b = Distance from axis of rotation of telescope to test target
L = Lateral deflection V = Vertical deflection E_v = Parallax correction

Note - The vertical coordinates of the test target should be corrected for telescope vertical parallax by adding algebraically the values of E_v to the values of V computed from the above tables. It is to be noted that the values of E_v are all positive and that this parallax correction is independent of the distance of the test target from the test fixture on which the sight is mounted.

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R014

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity (as indicated on reverse hereof).

SPECIFICATION

ORGANIZATION *(Of submitter)*

CITY AND STATE

CONTRACT NO.

QUANTITY OF ITEMS PROCURED

DOLLAR AMOUNT
$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*

DATE

DD FORM 1 APR 63 1426 REPLACES NAVSHIPS FORM 4863, WHICH IS OBSOLETE

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